DataCORE

Final Analysis, AI Results Check & Portfolio Prep

Afternoon: DataCORE

1. **Final Analysis + AI Results Check** **(1:00–1:45)**
2. **Final Dataset Documentation & Portfolio Prep** **(1:45–2:45)**
3. **Scientific Storytelling — Final Chart Set Selection** **(2:45–3:00)**

*Today you don't learn a new technique — you **finish.** One clean run, an honest error-check, and a portfolio you'd be proud to show at the Research Summit.*

1:00 – 1:45 · Final Analysis

Run it clean — exactly once

Before you trust a result, prove it **reproduces**:

1. **Restart** the kernel / session (clear memory).
2. **Run top to bottom** with no skipping.
3. Confirm **every number** still appears — no errors, no leftovers.

Every code block carries a one-line comment explaining its purpose — that's what makes it a portfolio piece.

> If it only works when you run cells **out of order**, it isn't a finished analysis — it's a lucky accident.

A clean, commented pipeline

Python R

```
wages = pd.read_csv("wages.csv")                      # load the raw data
wages = wages.dropna()                                # drop incomplete rows
gap = (wages.loc[wages.gender=="Male","hourly_wage"].mean()
       - wages.loc[wages.gender=="Female","hourly_wage"].mean())   # gender wage gap
r, p = stats.pearsonr(wages.experience_years, wages.hourly_wage)   # experience-wage link
print(f"N = {len(wages)}, gap = ${gap:.2f}, r = {r:.2f} (p = {p:.4f})")
```

```
N = 220, gap = $2.27, r = 0.49 (p = 0.0000)
```

Notice: **one comment per line of intent.** A stranger (or future you) can read it top to bottom and follow the logic.

AI results check — an honest second opinion

Paste your key results into the AI and ask:

"Are there any obvious errors or red flags in these results?"

Then compare its critique to **your own judgment.**

The AI is good at catching:

- Impossible values (a negative age, r > 1)
- Mismatched units or decimal slips
- Claims stronger than the numbers support

It can miss:

- Context only *you* know (how data was collected)
- Whether the test fits the question

Does it flag anything you missed? Does it miss anything important? You are still the analyst — the AI is the reviewer.

RcS — robustness: drop a control, re-run

Python | R

```python
full = smf.ols("hourly_wage ~ experience_years + C(education) + C(gender)",
               data=wages).fit()                                    # full model
reduced = smf.ols("hourly_wage ~ experience_years + C(gender)",
                  data=wages).fit()                                 # drop education
print(f"experience coef  (full)    = {full.params['experience_years']:.3f}")
```

```
experience coef  (full)    = 0.407
```

```python
print(f"experience coef  (reduced) = {reduced.params['experience_years']:.3f}")
```

```
experience coef  (reduced) = 0.426
```

Dropping **education** moves the experience coefficient by about **4%**. If a control swings your main coefficient a lot, that sensitivity **is a limitation** — document it honestly.

1:45 – 2:45 · Portfolio Prep

Your portfolio is an artifact, not a folder of files

Commented notebook
Every code block has a one-line comment explaining its purpose.

Exported figures
PNG, 300 DPI, descriptive file names.

Summary table
Formatted statistics table with an APA caption.

Data cleaning log
What you changed, why, and what you chose *not* to change.

AI usage log
Tools used, prompts that worked, errors the AI made.

Exported PDF
The notebook as a PDF — what you display at the Summit.

Summary statistics table + caption

Python R

```
summary = wages.groupby("education")["hourly_wage"].agg(
    ["count", "mean", "median", "std"]).round(2)    # summary stats by group
print(summary)
```

```
             count   mean  median   std
education
Associate       45  24.41   23.53  5.04
Bachelor        80  29.24   29.10  4.05
Graduate        27  34.51   34.98  4.07
High School     68  22.32   22.22  4.43
```

Table 1. *Hourly wage by education level (N = 220).* **Note.** Data from the 2023 wage survey; values in U.S. dollars per hour.

Two logs that make you credible

Data cleaning log

Step	Why
Dropped 6 blank rows	No wage recorded
Recoded "M/F" → gender	Consistent labels
Kept 3 high wages	Real, not errors

AI usage log

Prompt	Outcome
"Explain r = 0.42"	Useful plain-English
"Write my groupby"	Worked after 1 fix
"Is this causal?"	Over-claimed — corrected

Logging what you changed **and** how you used AI is honesty, not bureaucracy — it lets anyone retrace your steps.

Export figures & notebook

Figure · Python | Figure · R | Notebook PDF

```
plt.savefig("fig1_experience_wage_scatter.png",
            dpi=300, bbox_inches="tight")     # 300 DPI, descriptive name
```

The **exported PDF** is the deliverable you display at the Research Summit — code, output, figures, and your comments in one clean document.

2:45 – 3:00 · Scientific Storytelling

Lock your 2–3 Summit charts

For each candidate chart, ask three questions. If it fails one — **simplify or cut it.**

1. Relevant?

Does it **directly answer** your research question?

2. Readable?

Is it legible at **presentation size**, from the back of the room?

3. Explainable?

Can you explain it in **under 30 seconds**?

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A clear, simple chart beats a complex one every time. When in doubt, cut a variable, not a viewer.

15:00